P1 Ex. 7.1 $\chi(\omega) = \int_{-\infty}^{\infty} d\tau\, G(\tau)\, e^{i\omega\tau}$

(a) $G(\tau) = \chi_0\, \delta(\tau) \Rightarrow \chi(\omega) = \chi_0 \int_{-\infty}^{\infty} d\tau\, \delta(\tau)\, e^{i\omega\tau} = \chi_0$

(b) $G(\tau) = \chi_0\, \theta(\tau)\, e^{-\alpha\tau}$

$$\chi(\omega) = \chi_0 \int_0^{\infty} d\tau\, e^{-\alpha\tau + i\omega\tau} = \frac{\chi_0}{\alpha - i\omega} =$$

$$= \frac{\chi_0}{-i(\omega + i\alpha)} = \frac{i\chi_0}{\omega + i\alpha}$$

Pole at $\omega = -i\alpha$
$\Rightarrow$ regular in "ω" > 0 half-plane.

(c) $G(\tau) = \chi_0\, \theta(\tau)$

$$\Rightarrow \chi(\omega) = \chi_0 \int_0^{\infty} dt\, e^{i\omega t} \underbrace{e^{-\varepsilon t}}_{} = \frac{\chi_0}{\varepsilon - i\omega}$$

need this in order to make integral convergent
$\varepsilon = +0$

So the functional form is the same as in (b) but $\alpha \to \varepsilon = +0$

Hence $\chi(\omega) = \frac{i\chi_0}{\omega + i\varepsilon} = i\chi_0 \left(P\frac{1}{\omega} - i\pi\delta(\omega) \right) =$

$$= \chi_0 \left(P\frac{i}{\omega} + \pi\delta(\omega) \right)$$

(d) $G(\tau) = \chi_0\, \theta(\tau)\, \sin(\omega_0 \tau)$

$$\chi(\omega) = \chi_0 \int_0^{\infty} d\tau\, e^{i\omega\tau}\, \frac{e^{i\omega_0\tau} - e^{-i\omega_0\tau}}{2i} =$$

$$= \frac{\chi_0}{2i} \int_0^{\infty} d\tau \left(e^{i(\omega+\omega_0)\tau} - e^{i(\omega-\omega_0)\tau} \right) = \frac{\chi_0}{2i} \left(\frac{1}{\varepsilon - i(\omega+\omega_0)} - \frac{1}{\varepsilon - i(\omega-\omega_0)} \right)$$

$e^{-\varepsilon\tau}$ for convergence again

$$\chi(\omega) = \frac{\chi_0}{2}\left(\frac{1}{\omega+\omega_0+i\varepsilon} - \frac{1}{\omega-\omega_0+i\varepsilon}\right) = (\varepsilon \to +0) =$$

$$= \frac{\chi_0}{2}(-i\pi)\left[\delta(\omega+\omega_0) - \delta(\omega-\omega_0)\right] + \frac{\chi_0}{2}\left(P\frac{1}{\omega+\omega_0} - P\frac{1}{\omega-\omega_0}\right)$$

$$= \chi_0 \frac{i\pi}{2}\left(\delta(\omega-\omega_0) - \delta(\omega+\omega_0)\right) - \underbrace{\frac{\chi_0\omega_0}{\omega^2-\omega_0^2}}_{\text{in the sense of Principal value.}}$$

Note that $Re(\chi(\omega))$ = even function of frequency.

$Im(\chi(\omega))$ = odd function

P2

$$\varepsilon(\omega) = \varepsilon_0\left(1+\chi_e(\omega)\right), \quad \varepsilon'' = \varepsilon_0\,\chi_e'' = \frac{\beta_0\omega\tau}{1+\omega^2\tau^2}.$$

Kramers-Kronig $\quad \chi'(\omega) = \frac{2}{\pi} P\int_0^\infty \frac{x\,\chi''(x)}{x^2-\omega^2}\,dx$

$$\chi'(\omega) = \frac{2}{\pi}\,\frac{\beta_0}{\varepsilon_0}\,P\int_0^\infty dx\; x\,\frac{x\tau}{1+x^2\tau^2}\,\frac{1}{x^2-\omega^2} = \begin{pmatrix} y = x\tau \\ dx = \frac{dy}{\tau}\end{pmatrix}$$

$$= \frac{2\beta_0}{\pi\varepsilon_0}\,P\int_0^\infty \frac{dy}{\tau}\,\frac{y^2}{\tau(1+y^2)}\,\frac{\tau^2}{y^2-\omega^2\tau^2} =$$

$$= \frac{2\beta_0}{\pi\varepsilon_0}\,P\int_0^\infty dy\,\frac{y^2}{1+y^2}\,\frac{1}{y^2-\omega^2\tau^2} = \left(\text{denote } \omega^2\tau^2 = u^2\right)$$

$$= \frac{2\beta_0}{\pi\varepsilon_0}\,P\int_0^\infty dy\left(\frac{1}{1+y^2} + \frac{u^2}{y^2-u^2}\right)\frac{1}{1+u^2}$$

$$\frac{y^2}{(1+y^2)(y^2-u^2)} = \left(\frac{1}{1+y^2} + \frac{u^2}{y^2-u^2}\right)\frac{1}{1+u^2}.$$

$$I_1 = P\int_0^\infty dy\,\frac{1}{1+y^2} = \frac{\pi}{2}$$ (Principal part "P" does not play a role since the integral is regular)

$$I_2 = P\int_0^\infty \frac{dy}{y^2-u^2} = (y = ux) = \frac{1}{u}\,P\int_0^\infty dx\,\frac{1}{x^2-1} = \frac{1}{u}\int_0^\infty dx\,\frac{1}{2}\left(\frac{1}{x-1} - \frac{1}{x+1}\right)$$

$$= \frac{1}{2u}\left\{\int_0^{1-\varepsilon}\frac{dx}{x-1} + \int_{1+\varepsilon}^{L}\frac{dx}{x-1} - \int_0^{L}\frac{dx}{x+1}\right\}.$$

At the end we need to take limits: $\varepsilon \to +0$, $L \to \infty$

$$\int_0^{1-\varepsilon}\frac{dx}{x-1} = (x = 1-z) = \int_1^{\varepsilon}\frac{-dz}{-z} = -\int_\varepsilon^1 d(\ln z) = \ln(\varepsilon)$$

$$\int_{1+\varepsilon}^{L} \frac{dx}{x-1} = (x = 1+z) = \int_{\varepsilon}^{L-1} \frac{dz}{z} = \ln \frac{L-1}{\varepsilon}.$$

$$\int_{0}^{L} \frac{dx}{x+1} = \ln\left(\frac{L+1}{1}\right) = \ln(L+1)$$

$$\Rightarrow I_2 = \frac{1}{2u}\left\{\ln(\varepsilon) + \ln\left(\frac{L-1}{\varepsilon}\right) - \ln(L+1)\right\} = (\varepsilon \text{ drops out!})$$

$$= \frac{1}{2u} \ln\left(\frac{L-1}{L+1}\right) \xrightarrow[L\to\infty]{} 0.$$

Thus $\chi'(\omega) = \frac{2\beta_0}{\pi\varepsilon_0} \frac{\pi}{2} \frac{1}{1+(\omega\tau)^2} = \frac{\beta_0}{\varepsilon_0(1+\omega^2\tau^2)}$

$$\chi(\omega) = \chi' + i\chi'' = \frac{\beta_0}{\varepsilon_0(1+\omega^2\tau^2)}(1+i\omega\tau) =$$

$$= \frac{\beta_0(1+i\omega\tau)}{\varepsilon_0(1+i\omega\tau)(1-i\omega\tau)} = \frac{\beta_0}{\varepsilon_0(1-i\omega\tau)}$$

$$\Rightarrow \varepsilon(\omega) = \varepsilon_0 \chi(\omega) = \frac{\beta_0}{1-i\omega\tau}.$$

P3

Equation of motion

$$\ddot{\vec{r}} = -\omega_0^2 \vec{r} + \frac{q}{m} \dot{\vec{r}} \times \vec{B}_0 + \frac{q}{m} \vec{E}_\alpha(z,t)$$

$$\dot{\vec{r}} \times \vec{B}_0 = \hat{x}\, \dot{y} B_0 - \hat{y}\, \dot{x} B_0$$

Since wavelength $\lambda \gg$ typical r, we can neglect z-dependence of $\vec{E}_\alpha \to \vec{E}_0 \left[\cos(\omega t)\hat{x} + \cos(-\omega t + \alpha\frac{\pi}{2})\hat{y}\right]$

$\cos(\omega t - \alpha \frac{\pi}{2}) = \sin(\omega t) \sin(\alpha \frac{\pi}{2}) = \alpha \sin(\omega t)$.

for $\alpha = \pm 1$.

So $\vec{E}_\alpha(t) = \vec{E}_0 \left[\cos(\omega t)\hat{x} + \alpha \sin(\omega t)\hat{y}\right]$.

(1) $\ddot{x} - \frac{qB_0}{m}\dot{y} + \omega_0^2 x = \frac{q}{m} E_0 \cos(\omega t)$

(2) $\ddot{y} + \frac{qB_0}{m}\dot{x} + \omega_0^2 y = \frac{q}{m} E_0\, \alpha \sin(\omega t)$

Consider circular variables: $\begin{cases} \xi = x + iy \\ \eta = x - iy \end{cases}$

Multiply (2) by (i) and add to (1):

$$\ddot{\xi} - \frac{qB_0}{m}\underbrace{(\dot{y} - i\dot{x})}_{(-i)(\dot{x}+i\dot{y})} + \omega_0^2 \xi = \frac{qE_0}{m}\underbrace{\left(\cos(\omega t) + i\alpha \sin(\omega t)\right)}_{e^{i\alpha\omega t} \text{ for } \alpha = \pm 1}$$

$$\Rightarrow \ddot{\xi} + i\frac{qB_0}{m}\dot{\xi} + \omega_0^2 \xi = \frac{qE_0}{m} e^{i\alpha\omega t}$$

Look for solution $\xi(t) = \xi_0 e^{i\omega_\alpha t}$

$$\Rightarrow \dot{\xi} = i\omega_\alpha \xi, \quad \ddot{\xi} = -\omega_\alpha^2 \xi$$

$$\left[-\omega_\alpha^2 + i\frac{qB_0}{m}(i\omega_\alpha) + \omega_0^2\right]\xi_0 e^{i\omega_\alpha t} = \frac{qE_0}{m}e^{i\alpha\omega t}$$

$$\Rightarrow \omega_\alpha = \alpha\omega \quad \text{and}$$

$$\xi_0 = \frac{qE_0/m}{\omega_0^2 - \omega^2 - \alpha\frac{qB_0}{m}\omega}$$

Note that $\frac{qB_0}{m} = \Omega_c$ = cyclotron frequency

Thus for $\underline{\alpha = +1}$:

$$\xi(t) = x(t) + iy(t) = \frac{qE_0/m}{\omega_0^2 - \omega^2 - \Omega_c\omega}e^{i\omega t}$$

$\underline{\alpha = -1}$:

$$x(t) + iy(t) = \frac{qE_0/m}{\omega_0^2 - \omega^2 + \Omega_c\omega}e^{-i\omega t}$$

$\underline{\alpha = +1}$: $$x(t) = \mathrm{Re}(\xi(t)) = \frac{qE_0\cos\omega t}{m(\omega_0^2 - \omega^2 - \Omega_c\omega)}$$

$$y(t) = \mathrm{Im}(\xi(t)) = \frac{qE_0\sin\omega t}{m(\omega_0^2 - \omega^2 - \Omega_c\omega)}$$

$\underline{\alpha = -1}$: $$x(t) = \frac{qE_0\cos\omega t}{m(\omega_0^2 - \omega^2 + \Omega_c\omega)}$$

$$y(t) = \frac{qE_0(-\sin\omega t)}{m(\omega_0^2 - \omega^2 + \Omega_c\omega)}$$

$$\vec{r}(t) = x(t)\,\hat{x} + y(t)\,\hat{y}$$

$$\vec{p}(t) = q\,\vec{r}(t)$$

$\underline{\alpha = +1}$

$$\vec{p}(t) = \frac{q^2 E_0}{m(\omega_0^2 - \omega^2 - \Omega_c \omega)}\left(\cos(\omega t)\hat{x} + \sin(\omega t)\hat{y}\right)$$

$$= \frac{q^2/m}{\omega_0^2 - \omega^2 - \Omega_c \omega}\,\vec{E}_{\alpha=+1}(t)$$

$$\vec{P} = \underset{\substack{\uparrow \\ \text{concentration of atoms}}}{N}\,\vec{p} = \varepsilon_0 \chi \vec{E}$$

$$\Rightarrow \quad \chi_{\alpha=+1}(\omega) = \frac{Nq^2}{m(\omega_0^2 - \omega^2 - \Omega_c \omega)}$$

$\underline{\alpha = -1}$

$$\vec{p}(t) = \frac{q^2 E_0}{m(\omega_0^2 - \omega^2 + \Omega_c \omega)}\left(\cos(\omega t)\hat{x} - \sin(\omega t)\hat{y}\right)$$

$$= \frac{q^2/m}{\omega_0^2 - \omega^2 + \Omega_c \omega}\,\vec{E}_{\alpha=-1}(t).$$

$$\chi_{\alpha=-1}(\omega) = \frac{Nq^2}{m(\omega_0^2 - \omega^2 + \Omega_c \omega)}$$

Re-write denominators: $\omega^2 \pm \Omega_c \omega = \left(\omega \pm \frac{\Omega_c}{2}\right)^2 - \left(\frac{\Omega_c}{2}\right)^2$

But $\frac{1}{2}\Omega_c = \frac{qB_0}{2m} = \Omega_L$ = Larmor frequency.

Both denominators can be written as

$$\omega_0^2 - \omega^2 \mp 2\Omega_L\omega = \omega_0^2 - \left((\omega \pm \Omega_L)^2 - \Omega_L^2\right) =$$

$$= \omega_0^2 + \Omega_L^2 - (\omega \pm \Omega_L)^2 \quad \text{where } (+) \text{ for } \alpha = +1.$$

$$(-) \text{ for } \alpha = -1$$

$$\Rightarrow \quad \chi_\alpha(\omega) = \frac{Nq^2/m}{\omega_0^2 + \Omega_L^2 - (\omega + \alpha\Omega_L)^2}$$

Resonance, when denominator $= 0$, takes place at frequency $(\omega_\alpha + \alpha\Omega_L)^2 = \omega_0^2 + \Omega_L^2$

$$\omega_\alpha = -\alpha\Omega_L + \sqrt{\omega_0^2 + \Omega_L^2}$$

$$\left\{ \begin{array}{ll} \underline{\alpha = +1} & \omega = -\Omega_L + \sqrt{\omega_0^2 + \Omega_L^2} \\ \underline{\alpha = -1} & \omega = \Omega_L + \sqrt{\omega_0^2 + \Omega_L^2} \end{array} \right\} \text{different for different polarizations } \alpha.$$

Let's do inverse Fourier for (a):

$$\chi(\omega) = \frac{i\chi_0}{\omega + i\alpha}$$

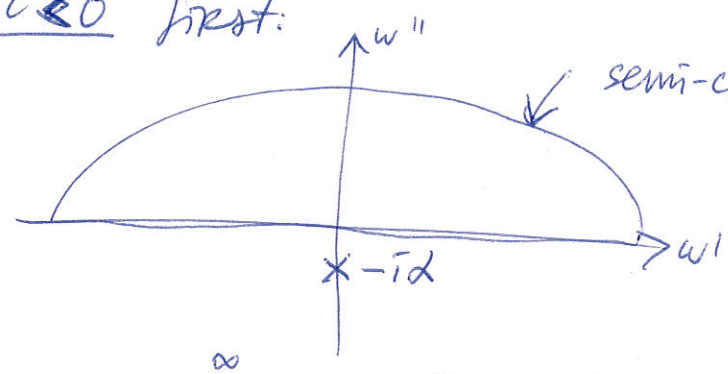
$$\chi(\tau) = G(\tau) = \int \frac{d\omega}{2\pi} \; e^{-i\omega\tau} \chi(\omega)$$

$$e^{-i\omega\tau} = e^{-i(\omega' + i\omega'')\tau} = e^{-i\omega'\tau + \omega''\tau} \Rightarrow$$

for $\begin{cases} \tau > 0 & \text{we need } \omega'' < 0 \\ \tau < 0 & \omega'' > 0 \end{cases}$

<u>Let $\tau < 0$</u> first:

$\tau = -|\tau|$

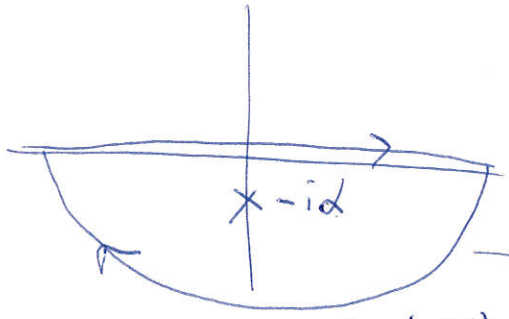


semi-circle gives nothing because $e^{-\omega''|\tau|} \to 0$, $\omega'' \to \infty$

$$\chi(\tau) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{e^{-i\omega\tau}(i\chi_0)}{\omega + i\alpha} = \frac{i\chi_0}{2\pi} \oint d\omega \frac{e^{-i\omega\tau}}{\omega + i\alpha}$$

$$= \frac{i\chi_0}{2\pi} 2\pi i \sum_{\substack{\text{residues} \\ \text{in } \omega''>0 \text{ plane}}} \left(\frac{e^{-i\omega\tau}}{\omega + i\alpha} \right) = 0$$ because there are no residues.

For <u>$\tau > 0$</u> close the contour in $\omega'' < 0$ half-plane:



$$\chi(\tau) = \frac{i\chi_0}{2\pi}(-2\pi i) \sum_{\text{res}} (\;) =$$

$$= \chi_0 e^{-\alpha\tau}$$

Residue $= e^{-i\tau(-i\alpha)} = e^{-\tau\alpha}$

$$\Rightarrow \boxed{\chi(\tau) = \chi_0 e^{-\alpha\tau} \theta(\tau)}$$